THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
FORM FOUR JOINT EXAMINATION FOR GOVERNMENT SCHOOLS –JEGOS

PRE-MOCK EXAMINATION
041 BASIC MATHEMATICS
MARKING SCHEME 2024

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1. (a) i. The numbers multiple of 5 are: 210, 6 015 ,12 300 ,1 330,
15 025,725, 5675……$00\frac{1}{2}$ mark

Their sum $= 210 + 6\,015 + 12\,300 + 1\,330 + 725 + 15\,025 + 5675$……$00\frac{1}{2}$ mark

$= 41\,280$ ……01 mark.

ii. The largest number is 74 976
The smallest number is 71

The difference$74\,976 - 71 = 74\,905$……01 mark

(b) The numbers are 9 , 7, and 5
Find the LCM and GCF 9, 7 , and 5……01 mark

3	9	7	5
3	3	7	5
5	1	7	5
7	1	7	1
	1	1	1……01 mark

The GCF is 1

The LCM $= 3 \times 3 \times 5 \times 7 = 315$ …… ..01 mark
The sum is $315 + 1 = 316$……01 mark

2. (a) $\frac{3}{3}[\log_2 x(x+1)] = \frac{3}{3}$

$\text{Log}_2(x^2 + x) = 1$

$x^2 + x = 2$ ……equation 1……01 mark

Solve equation 1 by factorization method
Product is-2, and factors are $(2, -1)$

$x^2 + 2x - x - 2 = 0$

$x(x+2) - 1(x+2) = 0$

$(x-1)(x+2)=0$

Either $x-1=0$ *or* $x+2=0$

$x=1$ *or* $x=-2$……02 marks

(b) $2^x \div 2^{-2y}=2^5$

$\frac{2^x}{2^{-2y}}=2^5$

$x+2y=5$ …… ……$00\frac{1}{2}$ mark

$2^{2x}\times 2^{2y}=2^4$

$2x+2y=4$

$x+y=2$ ……equation 2……$00\frac{1}{2}$ mark

Solve equation 1 and 2 by elimination method

$\begin{cases} x+2y=5 \\ x+y=2 \end{cases}$

$-\left|\begin{matrix} x+2y=5 \\ x+y=2 \end{matrix}\right.$

$y=3$……01 mark

Substitute 3 for$y$ *in equation* 2

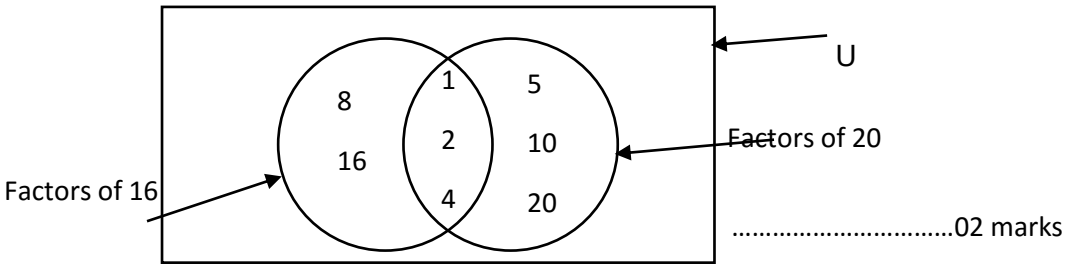
$x+y=2$

$x+3=2$

$x=2-3=-1$ ……01 mark

3. (a) i. Factors of 16 are 1, 2, 4, 8, and 16

Factors of 20 are 1, 2, 4, 5, 10, and 20



……02 marks

Figure 1

ii. The highest common factor of 16 and 20 is 4……01 mark

(b) Event (E)= $\{1,2,3,4,5\}, n(E)=5$……$00\frac{1}{2}$ mark

Sample space (S)= $\{1,2,3,..,20\}, n(S)=20$ ……$00\frac{1}{2}$ mark

P(E) = $\frac{n(E)}{n(S)}$ = $\frac{5}{20}$ = $\frac{1}{4}$……01 mark

Thus, the probability that all three numbers are less than or equal to 5 is

$P(E) \times P(E) \times P(E) = \frac{1}{4} \times \frac{1}{4} \times \frac{1}{4} = \frac{1}{64}$.....01 mark

4. (a) Express $2x - 3y = 5$ in form $y = mx + c$

$3y = 2x - 5$

$\frac{3y}{3} = \frac{2}{3}x - \frac{5}{3}$

$y = \frac{2}{3}x - \frac{5}{3}$

Compare $y = mx + c \; and \; y = \frac{2}{3}x - \frac{5}{3}$

$m = \frac{2}{3}$

i. Parallel lines case: $m_1 = m_2 = \frac{2}{3}$.....$00\frac{1}{2}$ mark

$\frac{2}{3} = \frac{y+1}{x-2}$

$2x - 4 = 3y + 3$

$2x - 4 - 3 = 3y - 3 + 3$

$3y = 2x - 7 \; \frac{3y}{3} = \frac{2x-7}{3}$

$y = \frac{2}{3}x - \frac{7}{3}$ .....01 mark

ii. Perpendicular line case:$m_1 m_2 = -1$

$\frac{2}{3} \; m_2 = -1 \; , \; m_2 = \frac{-3}{2}$.....$00\frac{1}{2}$ mark

$m = \frac{-3}{2}, (x_1, y_1) = (2, -1), (x, y)$

$\frac{-3}{2} = \frac{y+1}{x-2}$

$-3x + 6 = 2y + 2$

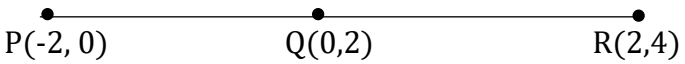
$-3x + 6 - 2 = 2y + 2 - 2$

$-3x - 4 = 2y$

$\frac{-3}{2}x + \frac{4}{2} = \frac{2y}{2}$

$y = \frac{-3}{2}x + 2$ ....01 mark

(b)



$\overline{PQ} + \overline{QR} = \overline{PR}$ .....00½ mark

d $= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$\overline{PQ} = \sqrt{(0 + 2)^2 + (2 - 0)^2}$ .....00½ mark

$= \sqrt{4 + 4}$

$= \sqrt{8}$ .....0½ mark

$\overline{PQ} = 2\sqrt{2}$ units

$\overline{QR}$

$= \sqrt{(2-0)^2 + (4-2)^2}$

$\overline{PQ} = \sqrt{4+4}$

$\overline{PQ} = \sqrt{8}$

$\overline{QR}$ $= 2\sqrt{2}$ units ……..00$\frac{1}{2}$mark

$\overline{PR} = \sqrt{(2+2)^2 + (4-0)^2}$

$= \sqrt{(4)^2 + (4)^2}$

$= \sqrt{16+16}$

$= \sqrt{32}$

$= 4\sqrt{2}$ ……..00½ mark

$\overline{PQ} + \overline{QR} = \overline{PR}$

$2\sqrt{2} + 2\sqrt{2} = 4\sqrt{2}$

LHS =RHS

Therefore points P (-2, 0), Q (0, 2), and R (2,4) are points on a straight line $\overline{PR}$ ..00$\frac{1}{2}$mark

5. (a) According to the diagram : $\angle A \cong \angle H \ and \ \angle C \cong \angle J$……..00$\frac{1}{2}$ mark

$\Delta ABC \sim \Delta HGJ$ by the Angle—Angle (AA) Similarity statement……..01 mark

Use the definition of similar triangles to find the value of x

Corresponding sides of similar triangles are proportional

Create and solve a proportional to find the value of$x$

$\frac{AB}{DE} = \frac{AC}{DF}$ Corresponding sides are proportional

$\frac{x+1}{3} = \frac{x+5}{6}$

$6x + 6 = 3x + 15$

$6x - 3x = 15 - 6$

$3x = 9, \frac{3x}{3} = \frac{9}{3}$

$x = 3$……01 mark

(b) The Side— $Side - Side$ (SSS) theorem used in proving congruence of polygons implies equality of corresponding sides while the $Side - Side - Side$ (SSS) theorem used in proving similarity of polygons means the ratio of corresponding edges or sides is constant or proportional……..03 marks

6. (a) Convert the weight of one rabbit into dg

$1\ kg = 10\ 000dg, 4\ kg = 40\ 000dg, 21\ dag = 2100dg$

Therefore $4\ kg\ 21\ dag\ 48\ dg = (40\ 000 + 2\ 100 + 48)$dg

$= 42\ 148\ dg$……01 mark

Again convert 50 kg 57 dag 76 dg into dg

$50\ kg = 500\ 000 dg, 57\ dag = 5\ 700\ dg$

Therefore 50 kg 57 dag 76dg $= (500\ 000 + 5\ 700 + 76) dg$

$= 505776\ dg$……01 mark

Number of rabbits $= \frac{505776\ dg}{42\ 148\ dg} = 12\ rabbits$ ……01 mark

(b). i. $I \propto pt\ i.e\ I = kpt\ where\ I = \$\ 43.75\ , P = \$\ 5000\ , t = \frac{1}{4}$

Therefore f$rom\ I = kpt$

$43\ .75 = k \times 5\ 000\ \times \frac{1}{4}$

$43.75 = 1250\ k$

$\frac{43.75}{1250} = \frac{1250\ k}{1250},\ k = 0.035$ ……..01 mark

Therefore $I = 0.035pt$……..01 mark

ii. $I = 0.035 \times 5\ 000 \times \frac{3}{4} = \$\ 131.25$……01 mark

7. (a) $75\% = 12\ 000\ 000\ shillings$ ……01 mark

$100\% = y$

$y = \frac{100\% \times 12\ 000\ 000\ shillings}{75\%}$……01 mark

$= 16\ 000\ 000\ shillings$……..01 mark

(b)

DR	CASH ACCOUNT							CR
Date	Particulars	Folio	Amount		Date	Particulars	Folio	Amount
1/1	Capital		500,000		6/1	Purchases		245,000
5/1	Sales		290,000		10/1	Purchases		204,000
29/1	Sales		320,000		14/1	Expenses		24,000
					31/1	Balance	c/d	637,000
			1,100,000					1,100,000
1/2	Balance	b/d	637,000					

DR	CAPITAL ACCOUNT						CR
Date	Particulars	Folio	Amount	Date	Particulars	Folio	Amount
31/1	Balance	c/d	500,000	1/1	Cash		500,000
				1/2	Balance	b/d	500,000

……03 marks

TOTAL BALANCE OF MR. PALM AS AT 31ST JANUARY 2020 IN TZS				
S/N	Account name	Folio	Dr	Cr
1.	Cash account		637,000	
2.	Capital account			500,000
3.	Sales account (290,000 + 320,000)			610,000
4.	Purchases (245,000 + 204,000)		449,000	
5.	Expenses account		24,000	
			1,110,000	1,110,000

8. (a) $S_n = \frac{n}{2}[2A_1 + (n-1)d]$

$S_{11} = \frac{11}{2}[2A_1 + (11-1)d]$

$S_{11} = \frac{11}{2}[2A_1 + 10\,d]$

$S_{11} = \frac{11}{2} \times 2[A_1 + 5d]$

$S_{11} = 11[A_1 + 5d]$ ……..01 mark

Middle term of 11 terms is given by $\left[\frac{n+1}{2}\right]^{th}$

$= \left[\frac{11+1}{2}\right]^{th} = \left[\frac{12}{2}\right]^{th} = 6^{th}$……..01 mark

Therefore A_6 is the middle term of the first 11 terms

$A_6 = A_1 + 5d = 130$

$S_{11} = 11 \times 130 = 1430$……..01 mark

(b) Let the first term be G_1 and the common ratio be r

From the information

$G_3 = G_1r^2 = 5$ ……..equation 1

$G_5 = G_1r^4 = 20$ ……..equation 2

Divide equation 1 into equation 2

$\frac{G_1r^4}{G_1r^2} = \frac{20}{5}$

$r^2 = 4,\ r = \pm 2$……01 mark

Substitute 2 for r >0 in equation 1

$G_1 2^2 = 5\,, 4G_1 = 5, \frac{5}{4} = G_1$……$00\frac{1}{2}$ mark

$For \quad r < 0,\ G_1(-2)^2 = 5\ \ 4G_1 = 5 = G_1 = \frac{5}{4}$ ……$00\frac{1}{2}$ mark.

The eighth term is given by $G_8 = G_1r^7$ for $r \geq 0$

$G_1r^7 = \frac{5}{4} \times 2^7$

$= \frac{5}{4} \times 128 = 160$ ……$00\frac{1}{2}$ mark.

For $r \leq 0, G_1r^7 = \frac{5}{4}(-2)^7 = -160$……$00\frac{1}{2}$ mark.

9. (a) . Pythagorean triple obeys $c^2 = c^2 + b^2$

$$a^2 = 10^2 + 24^2$$ .....01 mark

$$576 + 100 = c^2$$

$$676 = a^2$$

$$a^2 = 676$$.....01 mark

$$\pm\sqrt{676} = \sqrt{a^2}$$

$$a = \pm 26$$

$$a = 26, a \geq 0$$.....01 mark

(b)

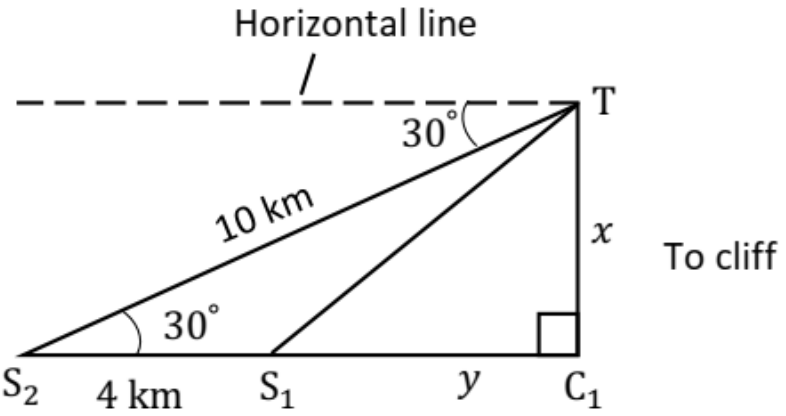


Figure 3

$$Sin30^\circ = \frac{opposite}{hypotenuse} = \frac{x}{10\ km}$$

$$x = 10\ km \times sin30^\circ$$

$$= 10\ km° \times 0.5 = 5km$$.....01 mark

Therefore the height of the cliff is 5 km

$$\cos 30^\circ = \frac{4+y}{10\ km}$$

$$y = 10\ km \times \cos 30^\circ$$

$$= 10\ km \times 0.8660$$

$$= 2.660\ km$$.....01 mark

$$TS_1C_1 = Tan\ \theta = \frac{opposite}{adjacent} = \frac{5}{2.660}$$

$$tan\theta = 1.8797, \theta = tan^{-1}(1.18797)\ , \theta = 61^\circ$$.....01 mark

10. (a) Let the numbers be $x\ and\ y$

$$x = \frac{1}{5}y$$

$x - \frac{1}{5}y = 0$ ……equation 1……$00\frac{1}{2}$ mark

$-x + y = 76$……equation 2……$00\frac{1}{2}$ mark

Solve equation 1 and 2 by elimination method

$$-\left|\begin{matrix} x - \frac{1}{5}y = 0 \\ -x + y = 76 \end{matrix}\right.$$

$\frac{4}{5}y = 76$

$\frac{4}{5}y \times \frac{5}{4} = 76 \times \frac{5}{4}$

$y = 95$ ……01 mark

Substitute 95 for y in equation 1

$x = \frac{1}{5} \times 95 = 19$ ……01 mark

(b) $5x^2 - 4x^2 - 55x + 40x = 60 - 6 = 0$

$x^2 - 15x + 54 = 0$ ……01 mark.

Solve the equation by factorization method

Product is 54 and factors are $(-9, -6)$ ……01 mark

$x^2 - 9x - 6x + 54 = 0$

$x(x - 9) - 6(x - 9) = 0$

$(x - 9)(x - 6) = 0$

Either $x = 6$ *or* $x = 9$

Therefore $x = 6$ *or* $x = 9$ ……01 mark

11. (a)

Class intervals	Cumulative frequency	frequency	Upper real class limit
$5 - 10$	7	7	10.5
$5 - 10$	15	8	16.5
$5 - 10$	25	10	22.5
$5 - 10$	38	13	28.5
$5 - 10$	47	9	34.5
$5 - 10$	50	3	40.5
Total		50……01 mark	…01 mark

Modal class is $23 - 28$

Mode $= l + \left[\frac{t_1}{t_1 + t_2}\right] c$

$= 22.5 + \left[\frac{13-10}{13-10+13-9}\right] 6$

$= 22.5 + \left[\frac{3}{3+4}\right] 6$……01 mark

$= 22.5 + \frac{36}{7}$

$= 22.5 + 5.1428 \ldots$

$= 27.64$ (2 decimal places)……01 mark

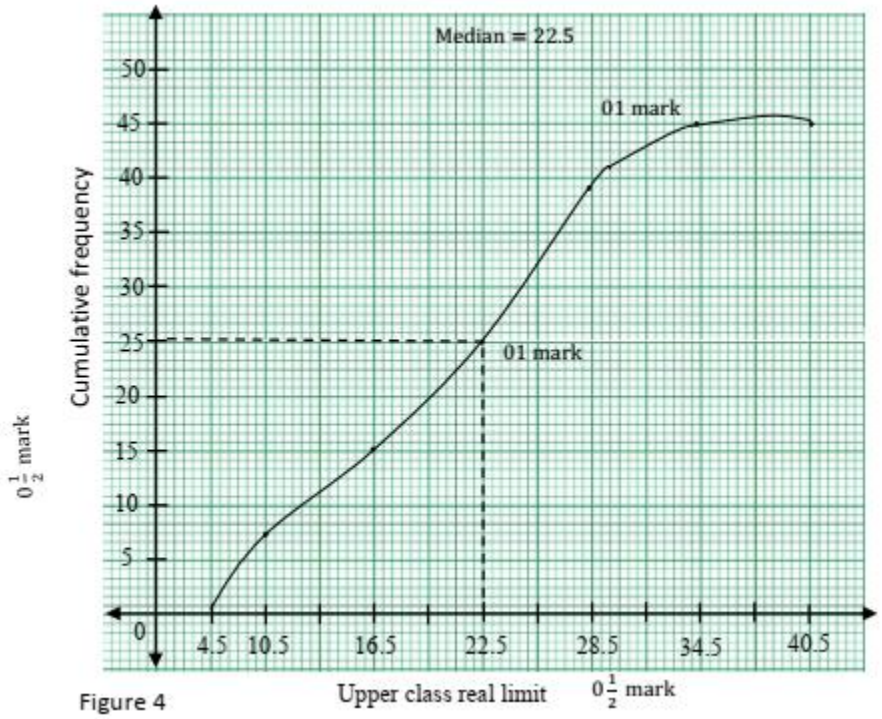


Figure 4

(b) $\overline{FH}$ *bisects* $\overline{GI}$

A segment passing through the centre bisects a tangent and form 90°
Therefore $4x + x = 90^{\circ}$

$5x = 90^{\circ}$, $\frac{5x}{5} = \frac{90^{\circ}}{5}, x = 18^{\circ}$……01 mark

$4x + x + 2x + \angle HGK = 180^{\circ}$(Straight line angle)

$7x + \angle HGK = 180^{\circ}$

$7 \times 18 + \angle HGK = 180^{\circ}$

$126^{\circ} + \angle HGK = 180^{\circ}$

$\angle HGK = 180^{\circ} - 126^{\circ}$

$\angle HGK = 54^{\circ}$ ……01 mark

$\angle HGK = \angle GJK = 54^{\circ}$(Angle between a chord and a tangent equals to the angle in the alternate segment)……01 mark

12. (a) i. $P\ and\ Q$ are on the same great circle or longitude

Calculate the angular distance $PQ = 39^{\circ} - 27^{\circ} = 12^{\circ}$……01 mark

Convert the angular distance into nautical miles

$1^{\circ} = 60\ nm$

$12^{\circ} = y$ ……01 mark

$y = \frac{12^{\circ} \times 60\ nm}{1^{\circ}} = 720\ nm$……01 mark

ii. $1nm = 1.852\ km$

$720\ nm = y$

$y = \frac{1.852\ km \times 720\ nm}{1nm}$……01 mark.

$= 1333\ km$……01 mark

(b) Volume of tank M= volume of tank Z

$V_M = l \times w \times h = V_Z = \pi r^2 h$ ……..01 mark

$l \times w \times h = \pi r^2 h$

$80\ cm \times 55\ cm \times 21\ cm = \frac{22}{7} \times r^2 \times 50\ cm$ ……..01 mark

$r^2 = \frac{80\ cm \times 55\ cm \times 21\ cm \times 7}{22 \times 50\ cm}$ ……$00\frac{1}{2}$ mark

$r^2 = 4 \times 4 \times 7 \times 7$……..01 mark

$\sqrt{r^2} = \sqrt{4 \times 4 \times 7 \times 7}$ ……..$00\frac{1}{2}$ mark

$r = 28\ cm$ …… …….01 mark

13. (a) Angular distance $= 20^{\circ} + 12^{\circ}$

$= 32^{\circ}$……01 mark

The distance is on small circle

Arc AB= $l_1 = \frac{\pi r \theta}{180^{\circ}}$ but $r = Rcos\ \alpha$

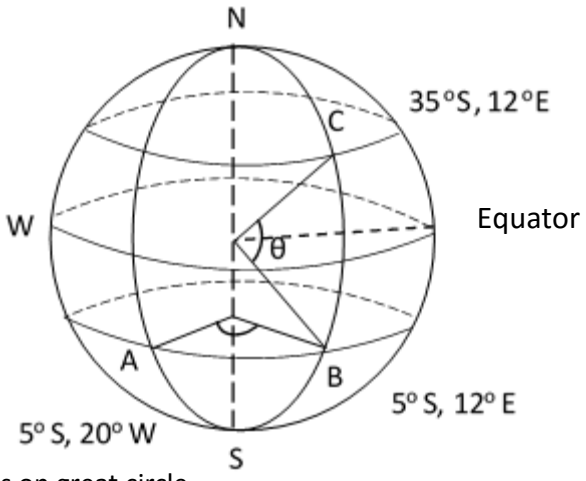
$Arc\ AB = l_1 = \frac{\pi \theta R\ Cos\ \alpha}{180^{\circ}}$

$= \frac{3.14 \times 6\,370\ km \times \cos 5^{\circ}}{180^{\circ}}$

$Arc\ AB = l_1 = 19\,925.69\ km$ ……..01 mark

Angular distance from point B to point C $= 5^{\circ} + 35^{\circ}$

$= 40^{\circ}$……01 mark



The distance is on great circle

Arc BC $l_2 = \frac{\pi r\theta}{180^\circ}$

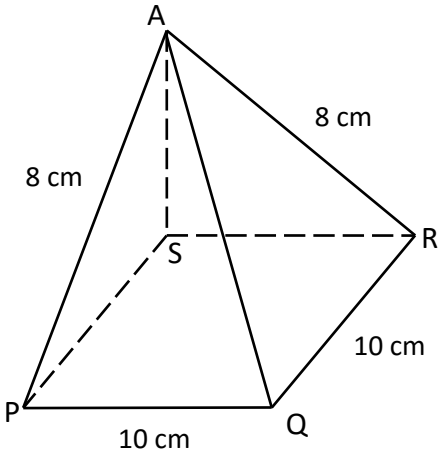
$Arc\ BC l_2 = \frac{3.14 \times 6\,370\ km \times 40^\circ}{180^\circ}$

$= 4\,444.84\ km$……01 mark

Total distance $Arc\ AB + Arc\ BC = l_1 + l_2 = 19\,925.69\ km + 4\,444.84\ km$

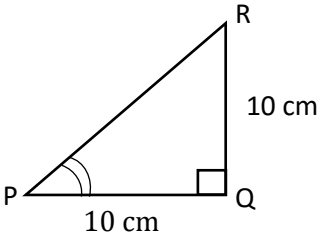
$= 24\,370.53\ km$ ……01 mark

(b)



Obtain $\overline{PR}$ from Δ PQR

By Pythagoras theorem:



$\overline{PR} = \sqrt{\left(\overline{PQ}\right)^2 + \left(\overline{RQ}\right)^2}$

$= \sqrt{10^2 + 10^2} = \sqrt{100 + 100}$

$= \sqrt{200}$

$= \left(10\sqrt{2}\right)cm$ ……01 mark

Let N be midpoint of $\overline{PR}$

$\overline{PN} = \frac{1}{2} \times \overline{PR} = \frac{1}{2} \times 10\sqrt{2} = (5\sqrt{2}\ cm$ ……01 mark

Height ($\overline{AN} = \sqrt{\left(\overline{AR}\right)^2 + \left(\overline{NR}\right)^2}$

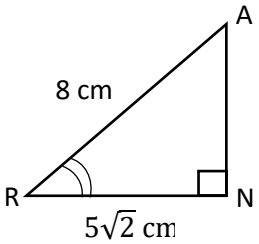
$= \sqrt{\left(\overline{AR}\right)^2 - \left(\overline{NR}\right)^2}$

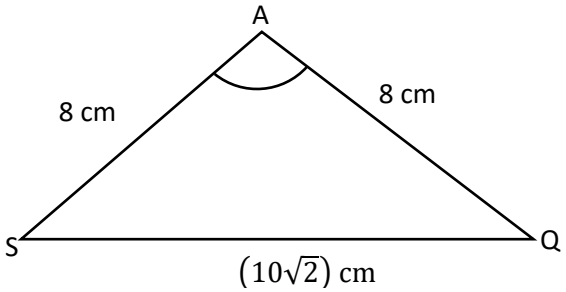
$= \sqrt{8^2 - \left(5\sqrt{2}\right)^2}$

$= \sqrt{64 - 50}$

$= \sqrt{14}$

$\approx 3.74\ cm$ ……01 mark





From cosine rule

$a^2 = b^2 + c^2 - bc\ Cos\ A$

$\left(10\sqrt{2}\right)^2 = 8^2 + 8^2 - 2 \times 8 \times 8 \times CosA$

$200 = 64 + 64 - 128\ Cos\ A$

$200 = 128 - 128\ Cos\ A$

$200 - 128 = -128\ Cos\ A$

$72 = -128 Cos\ A$

$\frac{72}{-128} = \frac{-128\ coa\ A}{-128}$

$Coa\ A = -0.5625$ ……$00\frac{1}{2}$ mark

$A = Cos^{-1}(-05625)$

$= 55^{\circ}46'$

$= 180^{\circ} + 55^{\circ}46'$

$= 235^{\circ}46'$

Therefore the angle AS makes with AQ is about $235^{\circ}\ 46'$……01 mark

14. (a) i. Table of values

$y = -3 + x$

x	-2	-1	0	1	2	3	4	5
$y = -3 + x$	-5	-4	-3	-2	-1	0	1	2

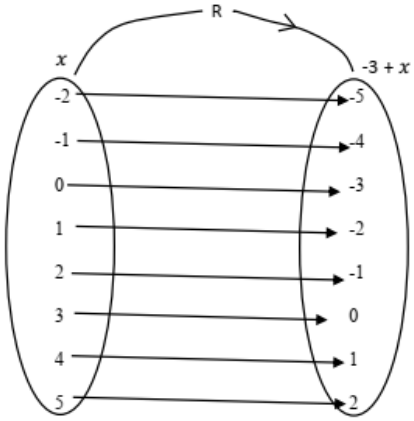
……..01 mark

Domain of R is $\{-2\ , -1, 0\ 1, 2, 3, 4, 5\}$……..01 mark

Range of R is $\{-5, -4, -3, -2, -1, 0, 1, 2\}$……01 mark

ii.

Figure 5



……02 marks

(b) Table of values

$y = x$

x	2	3	4	…
y	2	3	4	…

$y = 2$

x	-2	-1	0	1	2
$y = 2$	2	2	2	2	2

$y = x + 4$

x	…	-4	-3	-2
$y = x + 4$	…	0	1	2

……01 mark

Domain $of\ f = \{x: x \in \mathbb{R}\}$……01 mark

Range of $f = \{y: y \mathcal{E} \mathbb{R}\}$ ……01 mark

Figure 5

